

V2: Wellness Copilot for Colorectal Cancer Criteria

Overview

This page contains preliminary details on an advanced, v2 version of the Colorectal Cancer criteria cohorts. In this version, we will augment with Family History and parse free-text clinical (and maybe PDF) notes with LLMs. We could also look into using Athena Health's Quality Measures API to see what we are able to build with it.

Messaging Programs

I have written out four example messaging programs to demonstrate what could be achieved with the data that we have access to (and what data we think would be achievable). These are just suggestions and are pending input from strategy.

Program #2b: Screening Selection - At-Home Kits for Average Risk of Colorectal Cancer

Goal: For patients found to be in the numerator for Program 2a (due for colorectal cancer screening), if we have the data available indicating that this patient is at average risk for colorectal cancer, we could send the patient into a tailored program that lets them know that it appears they are eligible for an at-home screening test.

Note that patients must be at average risk for colorectal cancer to be eligible for at-home screening, see:

<https://www.cologuardhcp.com/about/patient-eligibility>

<https://www.bcbsmt.com/provider/education-and-reference/education/news-and-updates/2024-archive/07-02-2024-in-home-test-kits-for-colorectal-cancer-screening>

See the [definition of average risk of colorectal cancer](#) by the American Cancer Society.

Numerator:

- No personal history of colorectal cancer
- No personal history of adenomatous polyps
- No reported family medical history of colorectal cancer
- No personal history of familial adenomatous polyposis (FAP) or Lynch syndrome (hereditary nonpolyposis colon cancer or HNPCC)
- No personal history of inflammatory bowel disease (ulcerative colitis or Crohn's disease)
- No personal history of abdominal or pelvic radiation for a previous cancer
- No personal of abdominal or pelvic radiation for a previous cancer

Denominator:

- Found to be due for a colorectal cancer screening by the criteria in Program 2a.

Overview of Messaging:

A program could be created that adds tailored messaging letting the patient know it appears they are eligible for an at-home screening kit?

Data Dependencies:

All data dependencies for Program 2a.

Family Medical History.

Diagnosis/Condition data for personal medical history for high risk of colorectal cancer.

Procedure data for abdominal or pelvic radiation.

Frequency:

Logic runs monthly, but only message X times per year

Questions for Strategy:

Maybe we could reach out to Erin from PrimeCare and ask about the criteria she uses for messaging patients for Cologuard?

Level of Effort: MODERATE-HIGH

- We would need to gain access to family medical history, which we have not validated for any EHR today, but we should be to gain access to for all our major EHRs.
- We would need to find codes for all exclusion criteria and validate that we are able to find all the necessary data.

Program #2c: Screening Selection - Increased Risk of Colorectal Cancer ?

Goal: For patients found to be at increased risk of colorectal cancer, patients are not eligible for at home screening kits and colonoscopies are recommended. Colonoscopies may be recommended sooner than age 45 and more frequently than every 10 years. These recommendations are complex and depend on the patient's specific risk category and the [American Cancer Society states](#) "The American Cancer Society does not have screening guidelines specifically for people at increased or high risk of colorectal cancer. However, other professional medical organizations, such as the US Multi-Society Task Force on Colorectal Cancer (USMSTF), do put out such guidelines. These guidelines are complex and are best reviewed with your health care provider. In general, these guidelines put people into several groups (although the details depend on each person's specific risk factors)."

Numerator:

- Personal history of colorectal cancer OR
- Personal history of adenomatous polyps OR
- Reported family medical history of colorectal cancer OR
- Personal history of familial adenomatous polyposis (FAP) or Lynch syndrome (hereditary nonpolyposis colon cancer or HNPCC) OR
- Personal history of inflammatory bowel disease (ulcerative colitis or Crohn's disease) OR
- Personal history of abdominal or pelvic radiation for a previous cancer OR
- Personal of abdominal or pelvic radiation for a previous cancer

Denominator:

- Found to be due for a colorectal cancer screening by the criteria in Program 2a.
- Since Program 2a excludes patients with a history of colorectal cancer, these patients would need to be added back into the denominator here
- Possibly patients younger than 45 (maybe 35?) should be brought into this denominator too?

Overview of Messaging:

I'm unsure of what the messaging for this program would be, considering that the recommendations for screening depend on the patient's risk category and should be discussed with a healthcare provider.

Maybe an encouragement to discuss your personalized screening recommendations with your provider?

Data Dependencies:

All data dependencies for Program 2a.

Family Medical History.

Diagnosis/Condition data for personal medical history for high risk of colorectal cancer.

Procedure data for abdominal or pelvic radiation.

Frequency:

Logic runs monthly, but only message X times per year

Questions for Strategy:

I'm unsure if a specific messaging program for patients at high risk would be of value, considering that the recommendations for screening depend on the patient's risk category and should be discussed with a healthcare provider. Additionally, the "every 10 year" cadence for colonoscopies may not apply to the patient's unique risk category, but could at least be a starting place?

Level of Effort: MODERATE-HIGH

- We would need to gain access to family medical history, which we have not validated for any EHR today, but we should be to gain access to for all our major EHRs.
- We would need to find codes for all exclusion criteria and validate that we are able to find all the necessary data.

Program #5: Follow up from positive colonoscopy procedure

Goal: When a colonoscopy is performed, the performing physician removes any polyps found. Depending on the amount of polyps and the type of polyps, the patient is given a time frame with which to follow up with a repeat colonoscopy. This information should be available in free text format in the procedure report from the performing physician.

See: <https://gastro.org/clinical-guidance/follow-up-after-colonoscopy-and-polypectomy-a-consensus-update-by-the-u-s-multi-society-task-force-on-colorectal-cancer/>

<https://www.aafp.org/pubs/afp/issues/2021/0301/p314.html>

NOTE: This program should take precedence over Program #1. If a patient had a colonoscopy that found a quantity or type of polyp that increased the patient's risk stratification, the patient should be following up with a colonoscopy in a shorter time frame (not non-invasive stool screenings).

Numerator:

1. Patients who meet all of the following criteria
 - a. Diagnostic report from colonoscopy is available
 - b. Diagnostic colonoscopy report indicates a follow up time frame sooner than the standard 10 years
2. The patient is overdue for a colonoscopy, or will be due within 3 months (time frame?), based on the current date and the follow up time frame
3. The patient does not have an open referral to Gastroenterology for a colonoscopy

Denominator:

1. Patients with a visit in a primary care department within the clinic in the last 2 years (time frame configurable by customer)
2. Patients who have NOT had a total colectomy
3. Patients who have NOT been diagnosed with colorectal cancer

Overview of Messaging:

Your GI specialist recommends that you follow up with a repeat colonoscopy due to the polyps found on your last colonoscopy. Our records indicate that you are due to schedule your repeat colonoscopy. Call this number to schedule.

Data Dependencies:

Free text diagnostic reports from colonoscopy procedures

- Plus the infrastructure to analyze these free text reports and extract dates
- These reports could be in PDF format or could be a scanned image. Would we be able to use AI to convert these to text?

Referrals and Procedures data for colonoscopies.

Questions for Strategy:

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Level of Effort: HIGH

- Whether or not we will be able to reliably find the data that we need for colonoscopy reports is still unknown
- If we find the reports, they may be in PDF format (or a scanned image), requiring us to convert to free text
- We will need to extract the follow up date from free text (using AI?)

Data Availability

	Epic OCHIN	Epic non OCHIN	Nextgen	Athena Health	eCW	Practice Fusion
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Report from CT Colonography * *Due to the low availability of data and the rarity of the procedure, we recommend this remains out of scope	Maybe? Pending FHIR Access Possibly with the DocumentReference and Binary FHIR APIs	Maybe? Possibly with the DocumentReference and Binary FHIR APIs	Unclear Maybe this could be found in the DiagnosticReport or DocumentReference FHIR APIs?	Maybe? We could potentially find this in the FHIR DiagnosticReport API (could be in PDF format)	Possibly? Could be available with the Ellkay Get Patient Documents API https://lkcloud-api.readme.io/reference/post_kdocuments-getpatientdocuments	Possibly? This could potentially be available in the DocumentReference bulk FHIR API that Practice Fusion offers?
Report from Colonoscopy or Flex Sig	Maybe? Pending FHIR Access Possibly with the DocumentReference and Binary FHIR APIs	Maybe? Possibly with the DocumentReference and Binary FHIR APIs	Unclear Maybe this could be found in the DiagnosticReport or DocumentReference FHIR APIs?	Maybe? We could potentially find this in the FHIR DiagnosticReport API (could be in PDF format)	Possibly? Could be available with the Ellkay Get Patient Documents API	Possibly? This could potentially be available in the DocumentReference bulk FHIR API that Practice Fusion offers
Diagnosis /Condition data indicating high risk of colorectal cancer	YES (not validated yet)	YES (not validated yet)	YES (not validated yet)	YES (not validated yet)	YES (not validated yet)	YES (not validated yet)
Procedure data indicating high risk of colorectal cancer	YES (not validated yet)	YES (not validated yet)	YES (not validated yet)	YES (not validated yet)	YES (not validated yet)	YES (not validated yet)
Demographics	YES (validated) Caboodle PatientDim table	YES (pending validation) We should be able to access this data over the FHIR Patient endpoint (pending validation of performance).	YES (validated) The persons proprietary API has this	YES (validated) Patient FHIR resource	YES (pending validation) This will be in the Patient FHIR resource through Ellkay	YES (not validated yet) This is available in the Patient FHIR resource. Or we could retrieve from Ellkay.
Family History	Yes (pending validation) Although there is some family history data available in the DiagnosisEventFact table in Caboodle, much more is available in the FamilyMemberHistory FHIR API	YES (pending validation) We should be able to access this data over the FHIR FamilyMemberHistory endpoint (pending validation of performance).	YES (not validated yet) We should be able to find this in the chart/family-history proprietary API	YES (not validated yet) There is a Family History AthenaOne API we could use, we would just need to request access and re-certify	YES (pending validation) Should be able to find in FamilyMemberHistory FHIR resource through Ellkay	YES (not validated yet) If we were able to subsidize Ellkay for a Practice Fusion customer, we could likely fetch referrals from theFamilyMemberHistory FHIR API.
Quality Measures* *Unclear how useful this data will be	Maybe? Caboodle does have "BestPracticeAdvisories" tables. We could potentially find information on whether clinical alerts fired for a particular patient.	Maybe?	Maybe? There is the chart/alerts API, but the documentation does not include many details	YES (not validated yet) Athena actually has a Quality Measures API where we could search for a patient's quality measure for colorectal cancer and see whether it is Needs Work, Satisfied, Satisfied (but can be Due Soon), Not Satisfied (but can be Out of Range so fixable), Excluded, or Snoozed.	Maybe? There is the patient alerts API through Ellkay. This could contain clinical quality alerts.	Maybe? pending Ellkay build There is the patient alerts API. This could contain clinical quality alerts.